



HARIDWAR UNIVERSITY

Department of Computer Science & Engineering

QUESTION BANK

Course Code: 24BCS601

Course Title: CLOUD COMPUTING

Module 1: Introduction

Short type Questions:-

1. Explain the evolution of cloud computing from traditional computing systems.
2. State one key difference between traditional computing and cloud computing.
3. Summarize the purpose of OpenNebula in cloud environments.
4. State one function of Eucalyptus in cloud computing.
5. Explain the importance of simulation tools in cloud research.
6. State one reason for using CloudSim in cloud computing studies.
7. Explain the role of cloud service providers in cloud computing.
8. Name any two major cloud service providers.
9. Describe one benefit of cloud computing with a suitable usage scenario.
10. Mention one usage scenario where cloud computing is preferred over traditional computing.

Long type Questions:-

1. Define cloud computing and explain its key characteristics and advantages with suitable examples.
2. Discuss the evolution and history of cloud computing, highlighting major milestones and technological advancements.
3. Explain cloud computing architecture with a neat diagram, describing its main components.
4. Describe the different types of cloud deployment models and compare their features and use cases.
5. Explain various business models around cloud computing, including pay-as-you-go and subscription models.
6. Discuss the role of major cloud service providers and their contributions to the growth of cloud computing.
7. Explain the issues and challenges associated with cloud computing.
8. Describe open-source cloud platforms such as Eucalyptus, Nimbus, and OpenNebula.
9. Explain the role of CloudSim in cloud computing research and performance evaluation.
10. Discuss the risks involved in cloud computing and suggest possible mitigation strategies.



HARIDWAR UNIVERSITY

Module 2: Cloud Services

Short type Questions:-

1. What are cloud services?
2. What is Software as a Service (SaaS)? Give one example.
3. Mention any two features of SaaS.
4. What is Platform as a Service (PaaS)?
5. State two benefits of PaaS.
6. What is Infrastructure as a Service (IaaS)?
7. Name any two cloud service providers.
8. What is Database as a Service (DBaaS)?
9. List the different cloud deployment models.
10. Mention any two challenges in cloud adoption.

Long type Questions:-

1. Explain the different types of cloud services. Discuss SaaS, PaaS, and IaaS with their features and benefits.
2. Describe Software as a Service (SaaS). Explain its key features, benefits, and common use cases.
3. Explain Platform as a Service (PaaS) and discuss how it supports application development and deployment.
4. Discuss Infrastructure as a Service (IaaS) with suitable examples. Explain its advantages and limitations.
5. Explain emerging cloud services such as Database as a Service (DBaaS), Monitoring as a Service, and Communication as a Service.
6. Compare public, private, community, and hybrid cloud deployment models with respect to security, cost, and scalability.
7. Discuss the role of major cloud service providers such as Google, Amazon Web Services (AWS), Microsoft Azure, IBM, and Salesforce.
8. Explain the advantages of cloud computing for organizations and individuals.
9. Discuss the challenges and risks involved in cloud adoption and suggest possible solutions.
10. Explain how cloud service models and deployment models support modern business and enterprise applications.



HARIDWAR UNIVERSITY

Module 3: Collaborating Using Cloud Services

Short type Questions:-

1. Define cloud-based e-mail communication.
2. Explain CRM management in cloud services.
3. Name one cloud-based project management tool.
4. State the purpose of task management in cloud collaboration.
5. Explain the role of cloud calendars and scheduling tools.
6. Define cloud-based word processing tools.
7. Mention one benefit of cloud-based spreadsheet tools.
8. Define a cloud database.
9. Explain desktop services in cloud computing.
10. Describe workload management in cloud environments.

Long type Questions:-

1. Explain how cloud-based e-mail communication improves collaboration in organizations and educational institutions.
2. Discuss the role of Customer Relationship Management (CRM) systems delivered through the cloud.
3. Explain cloud-based project management tools and their importance in team collaboration.
4. Describe how cloud services support event management and task management.
5. Explain the use of cloud-based calendars and scheduling tools in collaborative environments.
6. Discuss the advantages of cloud-based word processing, presentation, and spreadsheet tools.
7. Explain how cloud databases enable real-time collaboration and data sharing.
8. Describe desktop services in the cloud and their benefits for organizations.
9. Discuss the role of social networks and groupware in cloud-based collaboration.
10. Explain workload management in cloud environments and its importance for efficient collaboration.

Module 4: Virtualization for Cloud

Short type Questions:-

1. Define virtualization.
2. State the need for virtualization in cloud computing.
3. List any two advantages of virtualization.
4. Mention one disadvantage of virtualization.
5. Name the different types of virtualization.



HARIDWAR UNIVERSITY

6. Define a system virtual machine.
7. Define a process virtual machine.
8. What is a Virtual Machine Monitor (VMM)?
9. Name any two virtualization platforms.
10. What is hardware virtualization?

Long type Questions:-

1. Explain the need for virtualization in cloud computing and discuss its advantages and disadvantages.
2. Describe the different types of virtualization used in cloud environments.
3. Explain system virtual machines and process virtual machines with suitable examples.
4. Discuss the role and functions of a Virtual Machine Monitor (VMM).
5. Explain the concept of machine properties in virtualization.
6. Describe interpretation and binary translation techniques used in virtualization.
7. Explain hardware virtualization and its importance in cloud computing.
8. Discuss the role of hypervisors and classify them with examples.
9. Compare Xen, KVM, VMware, VirtualBox, and Hyper-V as virtualization platforms.
10. Explain how virtualization forms the foundation of private and public cloud infrastructures.

Module 5: Security, Standards and Applications

Short type Questions:-

1. Define security in cloud computing.
2. Mention any two cloud security challenges.
3. Define SaaS security.
4. State the importance of cloud standards.
5. Name the role of the Open Cloud Consortium (OCC).
6. State the function of the Distributed Management Task Force (DMTF).
7. Mention one standard used for cloud security.
8. Explain end-user access in cloud computing.
9. State one security issue related to mobile internet devices in the cloud.
10. Mention one benefit of using standards in cloud applications.

Long type Questions:-



HARIDWAR UNIVERSITY

1. Explain the concept of security in cloud computing and discuss its importance in modern cloud environments.
2. Discuss the major cloud security challenges faced by organizations and users, and suggest suitable mitigation techniques.
3. Explain Software as a Service (SaaS) security and describe the shared responsibility model between the service provider and the user.
4. Describe cloud computing standards and explain why standardization is essential for interoperability and security.
5. Explain the role of the Open Cloud Consortium (OCC) in the development and promotion of cloud standards.
6. Discuss the objectives and contributions of the Distributed Management Task Force (DMTF) in cloud computing.
7. Explain the importance of standards for application developers, messaging, and security in cloud-based systems.
8. Describe end-user access mechanisms in cloud computing and the security measures required to protect user data.
9. Discuss the integration of mobile internet devices with cloud computing and the associated security concerns.
10. Explain how cloud security standards and best practices help organizations achieve secure and reliable cloud applications.